User interface design

Exp No. 2: - chunking model

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Instruction Page of the Memory Game effectively utilizes chunking as follows:

1. **Clear and Sequential Numbering**
   - The instructions are divided into **simple steps**, guiding users from viewing numbers to recalling them.
   - Each step presents **one key action**, reducing cognitive overload.
2. **Time Constraint Reinforcement**
   - Explicitly states the time limit: **"You will have 10 seconds to memorize the numbers."**
   - Prepares users to focus attention efficiently.
3. **Purpose of the Screen**
   - Visual memory encoding phase, where users view numbers and group them mentally.
   - Chunking principle helps users remember numbers in **smaller, meaningful units**.
4. **How the Chunking Phase Works**
   - Users scan numbers and create **mental chunks** (e.g., group them in pairs or patterns).
   - Timer limits viewing time, forcing quick memory encoding.
   - After time expires, users transition to the recall phase.

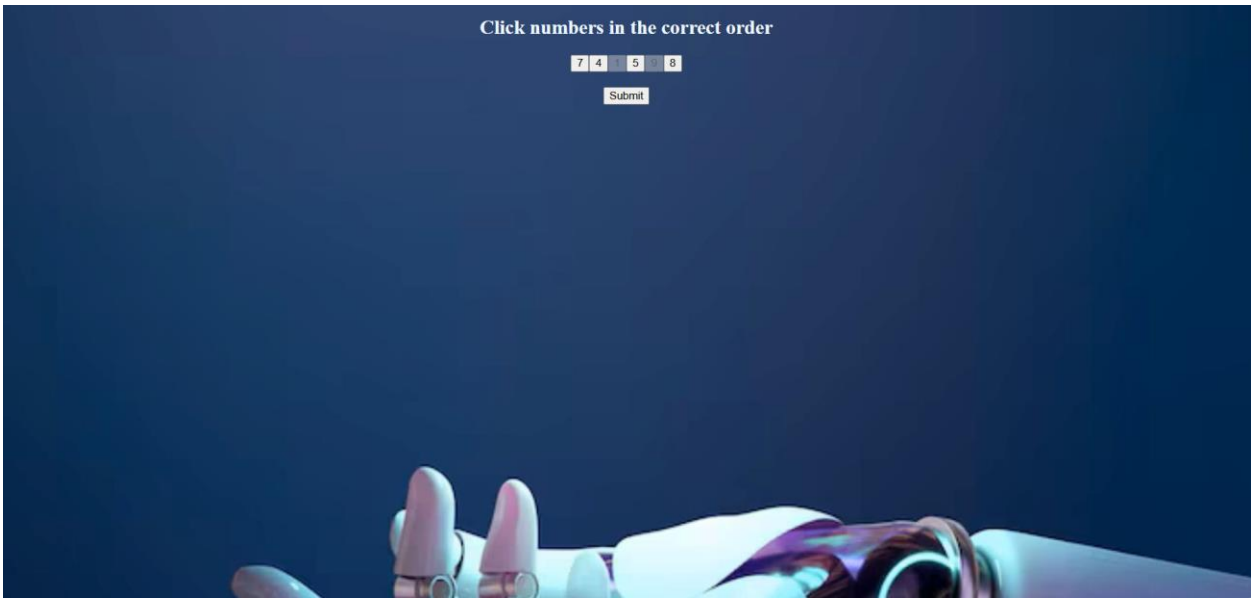
FRAME 2: RECALL PHASE (SELECTION PAGE)

Analysis of the Memory Game – Selection Phase Screen

This screen represents the recall phase, where users select the numbers in the correct order.

1. **Purpose of the Screen**
- Tests memory by asking users to recall the previously memorized numbers.
- Evaluates the effectiveness of chunking and short-term memory.
2. **Key Elements and UI Components**
- **Title & Instructions** – "Select numbers in the correct order."
  - Clear guidance for user action.
- **Shuffled Buttons of Numbers**
  - Displayed in a random order to increase difficulty.
  - Users must click numbers in the **original memorized sequence**.
- **Submit Button** – Confirms selections and moves to result page.
3. **How the Selection Phase Works**
- Users recall numbers from memory.
- They click the numbers sequentially according to the original order.
- Submission evaluates accuracy and stores the score.
4. **Cognitive and UX Benefits**
- Tests memory accuracy by comparing input with original sequence.

- Shuffling prevents simple guessing, improving cognitive challenge.
- Engaging interface with immediate feedback encourages active participation.



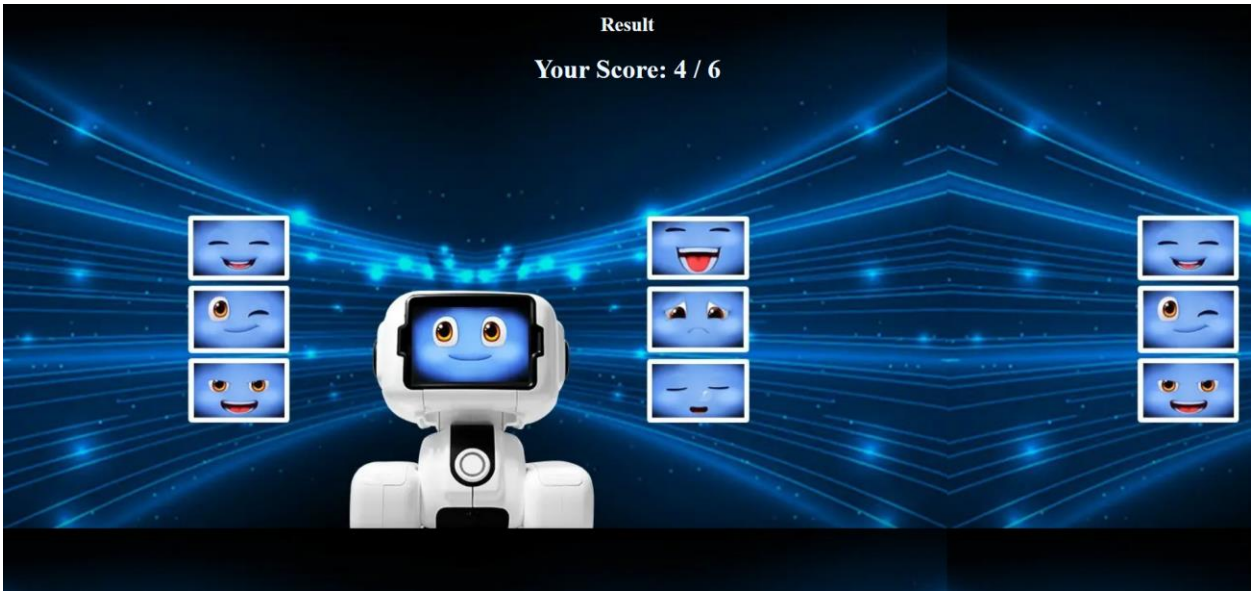
FRAME 3: RESULT PAGE (SCORE & FEEDBACK PAGE)

Analysis of the Memory Game – Score & Feedback Screen

This screen represents the score and feedback phase, where users receive performance evaluation.

1. **Purpose of the Screen**
- Provides **feedback on recall accuracy**.
- Allows users to **decide next action**: restart, continue, or exit.
2. **Key Elements and UI Components**
- **Title & Score Display**
  - Shows the user's score (e.g., 4/6 correct).
- **Action Buttons**
  - **Restart** – Retake the memory game.
  - **Exit** – Exit the game.
- **Bee Character (optional)** – Adds playful, gamified engagement.
3. **How This Phase Works**
- User's input from the Recall Phase is compared with the original sequence.

- Accuracy is displayed.
- Users choose the next action based on their performance.

4. Cognitive & UX Benefits

- Immediate feedback reinforces learning and memory strategies.
- Gives control to users over their learning process.
- Gamified elements make the task enjoyable and less stressful.